



Revitalizing the Archaeological Heritage of the Central Saharian Atlas: A GIS-Based Methodological Approach to Updating Stéphane Gsell's Atlas (Sheet No. 46 – Djelfa)

Jaballah Nacima*

University of Algiers 2 (Algeria)

nacima.jaballah@univ-alger2.dz

Rabhi Merouane

University of Algiers 2 (Algeria)

merouane.rabhi@univ-alger2.dz

Abstract ;

Adopting a systematic spatial analysis methodology, the research proceeded through the digitization and georeferencing of Gsell's original plates, correcting historical projection distortions to align with modern geodetic standards. This process facilitated the transmutation of qualitative, descriptive archival data into a dynamic, vector-based Spatial Database. By overlaying this historical dataset onto modern high-resolution Digital Elevation Models (DEM) and integrating it with recently documented archaeological inventories, the study investigates the dialectic relationship between ancient settlement patterns and the landscape. The results reveal a statistically significant correlation between Protohistoric human occupation and specific geomorphological characteristics. The spatial distribution analysis highlights a marked preference for altitudinal zones ranging between 1000m and 1400m, suggesting a settlement strategy driven by environmental adaptation, resource accessibility, and defensive monitoring. Furthermore, the research provides a critical comparative reading between the distribution of funerary monuments (specifically Bazinas and Dolmens) recorded by Gsell and the spatial clustering of rock art stations. This multi-layered approach not only validates the historical accuracy of Gsell's work but also enhances it, offering new insights into the cultural landscape of the Djelfa region during Protohistory.

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* author Corresponding

Introduction

The "Atlas archéologique de l'Algérie", compiled by the eminent historian Stéphane Gsell between 1902 and 1911, stands as the magnum opus of colonial archaeological cartography in North Africa. Comprising 51 sheets, this monumental inventory has served for over a century as the primary reference for identifying and locating ancient sites across the Maghreb (Gsell, 1911). Despite its undeniable historical value and the rigor of its initial compilation, the Atlas, in its traditional analog format, presents significant epistemological and technical challenges for contemporary research. The reliance on obsolete geodetic systems (referencing the Paris Meridian) and the static nature of paper-based documentation constitute a barrier to modern analytical approaches, which demand high-precision geolocation and the ability to correlate archaeological data with dynamic environmental variables (Wheatley and Gillings, 2002). [1]

In the context of the digitization of heritage, the transition from classical cartography to Geographic Information Systems (GIS) has become an imperative. This study focuses on the Djelfa region (Sheet No. 46), a territory strategically situated within the Central Saharan Atlas, specifically the Ouled Naïl range. Characterized by an exceptional density of archaeological remains—spanning from the Neolithic to the Protohistoric and

Roman periods—this region offers an ideal "laboratory" for applying the principles of Archaeomatics. The area's rich landscape of funerary monuments (tumuli, bazinas, dolmens) and rock art stations provides a complex dataset suitable for testing the efficacy of digital spatial analysis.

Consequently, this research addresses a dual problematic: Methodologically, how can modern GIS technologies be employed to rectify, georeference, and update the centenary data of Stéphane Gsell, thereby bridging the gap between historical archives and modern spatial databases? Interpretatively, how can the resulting high-resolution digital maps and Digital Elevation Models (DEMs) enhance our understanding of ancient settlement patterns, revealing the deterministic relationship between human occupation and the geomorphological landscape of the Djelfa region?

1 General Methodology and Cartographic Trans-formation Protocol

Methodology and Cartographic Transformation Protocol The operational framework of this study was predicated on a rigorous technical protocol designed to migrate archaeological data from a static, analog environment to a dynamic, digital ecosystem. To ensure the integrity of the spatial data extracted from Sheet No. 46 (Djelfa) of the Atlas archéologique de l'Algérie, a systematic three-stage workflow was implemented within a Geographic Information System (GIS) environment.

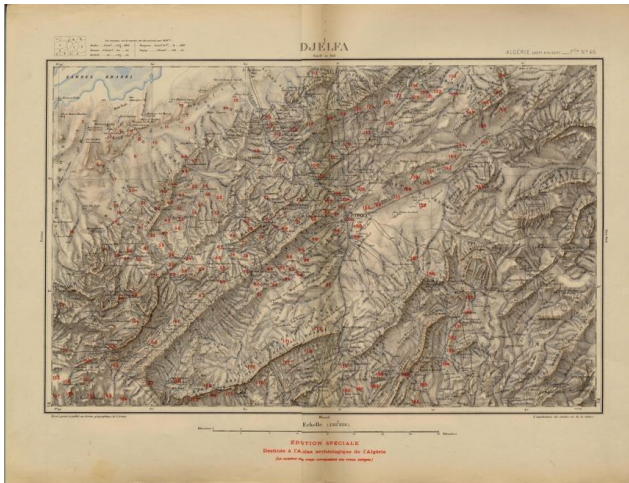


Figure 1: STEPHAN GSELL Djelfa region (Sheet No. 46), 1911

1.1 Data Acquisition and Pre-processing

The initial phase involved the high-resolution raster scanning of Stéphane Gsell's original lithographic plate. The map was digitized at a resolution of 300 dpi (dots per inch). This resolution was critical not only to preserve the legibility of minute typographic details and symbology but also to mitigate the effects of physical deterioration. Historical cartographic documents are frequently subject to mechanical wear and hygroscopic deformation (such as paper shrinkage or expansion folds), which can introduce significant geometric errors if not adequately addressed during the pre-processing stage (Conolly and Lake, 2006).

1.2 Georeferencing and Coordinate System Rectification

This phase constituted the technical core of the study. The original 1911 map was constructed using obsolete French colonial projections (often the Bonne Projection or the Voirol system), referencing the Paris Meridian rather than the Greenwich Prime Meridian used today. To bridge this geodetic gap and integrate the data with modern spatial infrastructures, the following procedures were executed using ArcGIS (v.10.x) and QGIS software:

1. Selection of Ground Control Points (GCPs): A network of spatially distributed Ground Control Points was established. These points were selected based on immutable topographic features visible on both the historical Gsell map and the modern 1:50,000 topographic base maps provided by the National Institute of Cartography and Remote Sensing (INC). The GCPs included triangulation vertices, major wadi confluences, and permanent colonial-era landmarks, ensuring a uniform distribution across the sheet to prevent local warping.

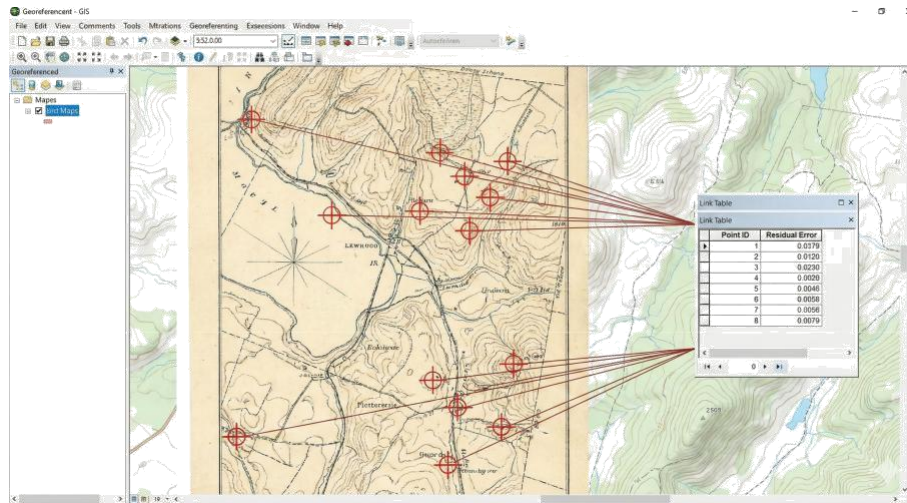


Figure 2: The Georeferencing Process. This image illustrates the creation of Ground Control Points (GCPs) in a GIS environment. The historical Gsell map (left) is linked to a modern topographic map (right) using common features, with the "Link Table" showing the residual errors for each point.

2. Geometric Transformation

Algorithm: Given the significant scale discrepancy

(1:200,000 vs. 1:50,000) and the non-linear physical distortions inherent in the century-old paper map, a simple affine transformation was deemed insufficient. Instead, a 2nd Order Polynomial Transformation was applied. This algorithm allows for "rubber-sheeting," effectively correcting complex, non-linear geometric distortions and curving the raster dataset to fit the modern coordinate grid (Wheatley and Gillings, 2002).

3. Projection Standardization: The dataset was ultimately re-projected into the World Geodetic System 1984 (WGS84), utilizing the Universal

Transverse Mercator (UTM) Zone 31N projection. This standardization ensures full interoperability with modern Global Positioning System (GPS) units for future fieldwork.

4. Accuracy Assessment (RMSE): The precision of the georeferencing process was validated by calculating the Root Mean Square Error (RMSE). The transformation was iteratively refined until the residual error was minimized to an acceptable threshold relative to the original map scale, ensuring that the spatial displacement of archaeological sites remained within statistically negligible limits.

Vectorization and Spatial Database Construction

Following the geometric stabilization of the raster map, the process transitioned to "Heads-up Digitization" (on-screen manual vectorization). This step involved the decomposition of Gsell's cartographic symbols into discrete digital layers:

1.Geometry Layer (Points):
Archaeological sites (Ruins, Bazinas, Dolmens) were

vectorized as point features in distinct Shapefiles, preserving their exact spatial location as depicted in the rectified map

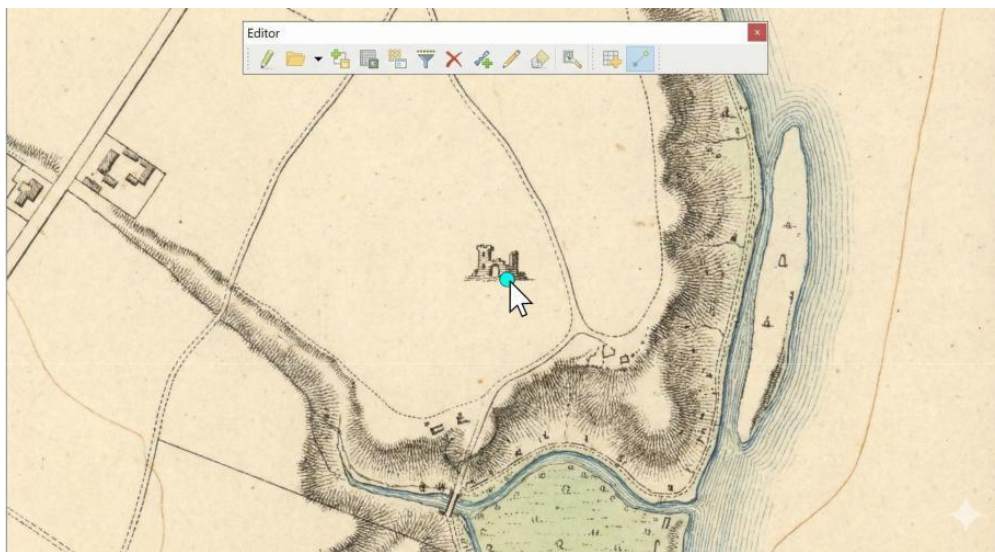
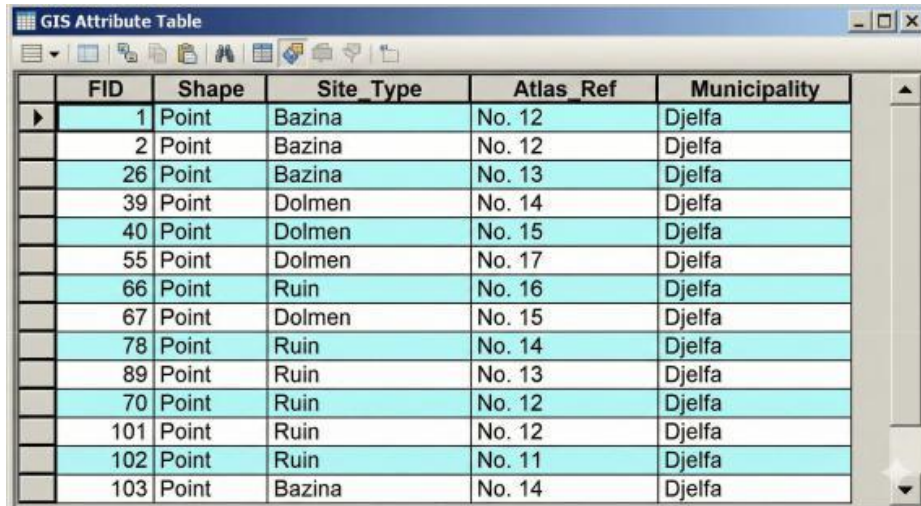


Figure 3: Heads-up Digitization. A vector point is created on top of a historical ruin symbol using the GIS editor tools.

2.Attribute Data Integration: Beyond mere spatial coordinates, the semantic information associated with each site (site type, atlas reference number, administrative location) was populated into the Attribute Table. This transformed the map from a passive visual document into a queryable Spatial Database, capable of supporting complex SQL queries and advanced statistical spatial analysis.



FID	Shape	Site Type	Atlas Ref	Municipality
1	Point	Bazina	No. 12	Djelfa
2	Point	Bazina	No. 12	Djelfa
26	Point	Bazina	No. 13	Djelfa
39	Point	Dolmen	No. 14	Djelfa
40	Point	Dolmen	No. 15	Djelfa
55	Point	Dolmen	No. 17	Djelfa
66	Point	Ruin	No. 16	Djelfa
67	Point	Dolmen	No. 15	Djelfa
78	Point	Ruin	No. 14	Djelfa
89	Point	Ruin	No. 13	Djelfa
70	Point	Ruin	No. 12	Djelfa
101	Point	Ruin	No. 12	Djelfa
102	Point	Ruin	No. 11	Djelfa
103	Point	Bazina	No. 14	Djelfa

Figure 4: Heads-up Digitization. A vector point is created on top of a historical ruin symbol using the GIS editor tools.

2 Our Methodology and Technical Protocol

The operational framework of this research is predicated on a rigorous, systematic technical protocol designed to migrate analog archaeological data into a dynamic Geographic Information System (GIS) environment. To realize the study's objectives, the workflow was structured into three fundamental, sequential phases, the outcomes of which are documented in the appended cartographic plates.

2.1 Phase I: Primary Data Acquisition and Georeferencing

The foundational stage involved the digital acquisition of (Stéphane Gsell's Atlas archéologique de l'Algérie), specifically Sheet N° 46 (Djelfa). The original lithographic plate was subjected to high-resolution raster scanning (e.g., bigger or equal to 300 dpi) to ensure the fidelity of

minute cartographic details and symbology.

A critical challenge in utilizing this historical document lies in its obsolete geodetic reference system. Produced at a scale of 1:200,000 at the turn of the 20th century, the map utilizes outdated colonial projections (often based on the Paris Meridian rather than the modern Greenwich standard) and suffers from physical deformations inherent to aged paper media. Consequently, direct integration with modern spatial data infrastructures is impossible without marked geometric discrepancies.

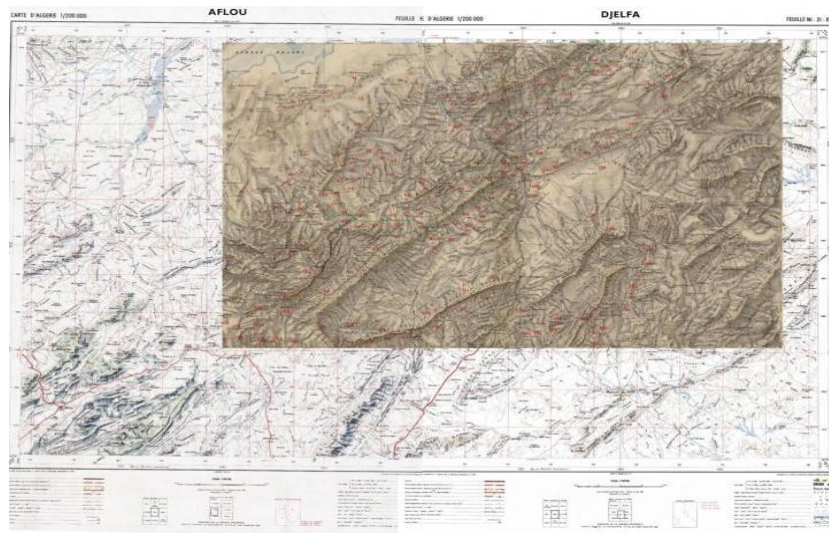


Figure 5: Rabhi Merouane's Superposition using Gsell's sheet n46.

To resolve these spatial inconsistencies, a rigorous geometric rectification (Georeferencing) procedure was executed within a GIS environment (e.g., ArcGIS or QGIS). This process entailed the establishment of a robust network of Ground Control Points (GCPs). These points were carefully selected based on immutable topographic features—such as prominent mountain peaks, distinct wadi confluences, or permanent colonial-era infrastructure—that are clearly identifiable on both the historical "source" map and modern, high-precision 1:50,000 topographic base maps (provided by the INC) serving as the "target" reference.

2.2Phase II: Vectorization and Spatial Database Architecture

Upon the successful completion of the georeferencing process, the methodology advanced to the digitization phase, marking

the transition from a passive raster environment to an active vector-based system. This phase involved the systematic transmutation of Gsell's cartographic semiology—specifically the red numeric indicators denoting ruins and sites—into discrete geospatial entities. 1.Heads-up Digitization and Feature Extraction :

The process utilized "heads-up digitization" (on-screen tracing), where the rectified map served as the background layer. Each archaeological occurrence identified on Gsell's Sheet No. 46 was manually plotted as a vector point feature. This vectorization process is critical as it isolates the archaeological data from the topographical noise of the original map, allowing for independent spatial manipulation (Wheatley and Gillings, 2002).

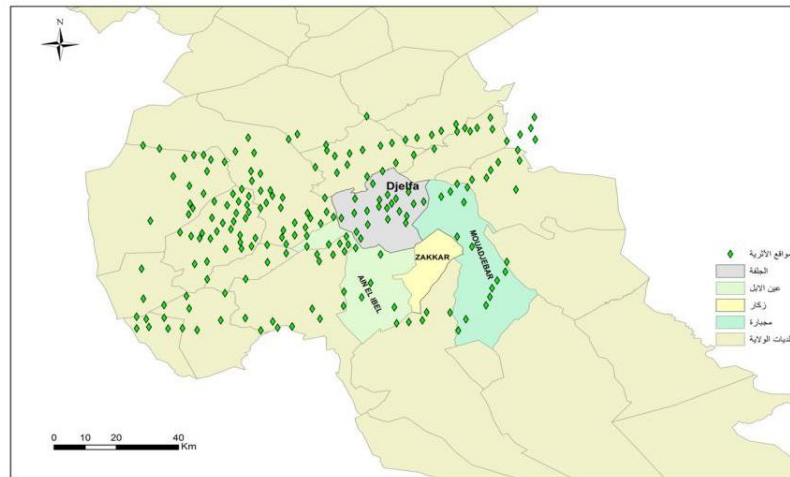


Figure 6: Rabhi Merouane's setting the protohistoric archaeological points on different zones of Djelfa

2. Structuring the Attribute Data :

Concurrently with the spatial plotting, a comprehensive Attribute Table was structured to house the semantic data associated with each vector point. Unlike the static paper map, the digital spatial database allows for multi-dimensional querying. The database schema was designed to include specific fields such as:

- 1) Site ID: A unique identifier corresponding to Gsell's original numbering.
- 2) Typology: A standardized classification of the features (e.g., Bazina, Dolmen, Tumulus, Rock Art Station).

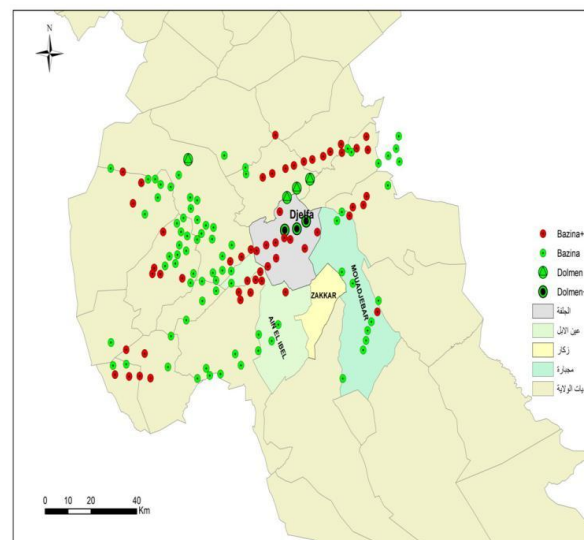


Figure 7: Rabhi Merouane's spatial distribution of protohistoric archaeological sites and their typology in the study area

3. Administrative Location: Through a "Spatial Join" operation, the archaeological points were overlaid with the modern

administrative boundary layers, automatically assigning each site to its respective municipality (Djelfa, Ain El Ibel, Zaccar, Moudjebara).

FID	Shape	Gsell_No	Site_Type	Municipality
1	Point	12	Bazina	Djelfa
12	Point	12	Bazina	Djelfa
20	Point	45	Dolmen	Ain El Ibel
40	Point	45	Dolmen	Ain El Ibel
50	Point	89	Bazina	Zaccar
60	Point	64	Bazina	Djelfa
70	Point	75	Ruin	Ain El Ibel
80	Point	86	Ruin	Djelfa
99	Point	97	Ruin	Ain El Ibel
100	Point	108	Ruin	Ain El Ibel
111	Point	111	Ruin	Ain El Ibel
122	Point	110	Ruin	Ain El Ibel
133	Point	131	Dolmen	Ain El Ibel
144	Point	132	Dolmen	Ain El Ibel
155	Point	143	Ruin	Ain El Ibel
176	Point	144	Ruin	Ain El Ibel
189	Point	155	Ruin	Ain El Ibel
190	Point	176	Rock Art	Zaccar
201	Point	177	Rock Art	Zaccar
212	Point	178	Rock Art	Zaccar
223	Point	199	Rock Art	Djelfa
223	Point	200	Rock Art	Zaccar
234	Point	231	Rock Art	Djelfa
235	Point	232	Ruin	Zaccar
256	Point	233	Ruin	Zaccar
267	Point	234	Ruin	Zaccar
268	Point	235	Rock Art	Zaccar
289	Point	236	Rock Art	Zaccar
290	Point	239	Dolmen	Ain El Ibel
291	Point	330	Dolmen	Ain El Ibel
392	Point	331	Ruin	Zaccar
303	Point	332	Dolmen	Zaccar

Figure 8: Structure of the Spatial Database. This visualization demonstrates the Attribute Table within the GIS environment. It highlights how the spatial points (Vector Data) are linked to descriptive information (Tabular Data), classifying sites by Gsell's reference

number, site type (Bazina, Dolmen, Ruin, Rock Art), and administrative municipality.

2.3 Phase III: Advanced Spatial Analysis and Terrain Model-ing

The final stage of the methodological protocol moved beyond simple inventorying to investigate the dialectic relationship between archaeological sites and their landscape context. To achieve this, a Digital Elevation Model (DEM)—likely derived from SRTM (Shuttle Radar Topography Mission) data with a 30-meter resolution—was integrated into the GIS environment.

This raster dataset served as the foundation for generating derived topographic surfaces, including Slope (Clinometry), Aspect (Orientation), and Hypsometric (Elevation) maps. By executing an "Overlay Analysis", the vector points representing archaeological sites were superimposed onto these environmental layers. This procedure, visualized in the appended Maps shown previously, allowed for the extraction of precise geomorphological variables for every site recorded by Gsell. This analytical step is pivotal for testing hypotheses regarding site selection strategies, resource accessibility, and

visibility during the Protohistoric period
(Conolly and Lake, 2006).

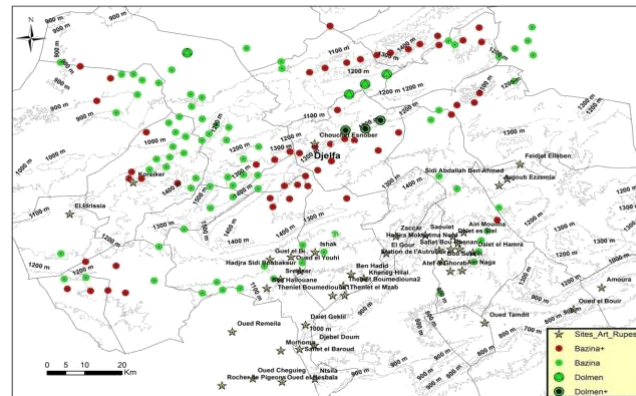


Figure 9: Topographic Analysis Concept - Rabhi Merouane's Proto-historical points applied in the map

Both of this visualizations illustrates the overlay of protohistoric archaeological sites (red points) onto a color-coded elevation model. This

technique, applied in the previous Maps allowing researchers to correlate settlement density with specific altitude zones and terrain relief (Bazina, Dolmen, Ruin, Rock Art)

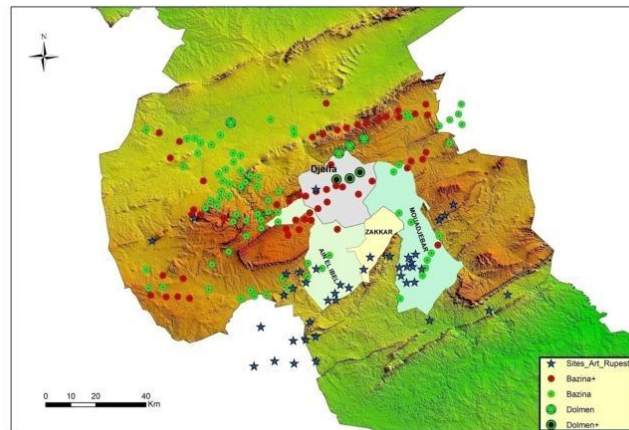


Figure 10: Rabhi Merouane's Digital Elevation Model (DEM) Analysis.

3 Results and Discussion :

3.1 Spatial Distribution and Zoning of the Funerary Land-scape:

The spatial interrogation of the digitized dataset—specifically the distribution patterns illustrated in Map reveals a

compelling narrative about the Proto-historic occupation of the Central Saharan Atlas. When we move beyond the raw data points and analyze the landscape context, a distinct "spatial dichotomy" emerges regarding the placement of funerary architecture.

The Logic of Agglomeration:

The distribution of Bazinas and Dolmens is far from being a stochastic or random scattering of stones. Instead, the updated map reveals a deliberate, highly structured pattern of clustering. These monuments appear in dense agglomerations, particularly along the mountainous piedmonts and the southern slopes of the Djelfa and Zaccar municipalities. This phenomenon suggests that these sites were not merely isolated burials but organized necropolis complexes, serving specific communities or clans over extended periods.

Topographic Determinism and Strategic Control:

By cross-referencing the site locations with the underlying topographic variables (as corrected from Gsell's Atlas), a clear settlement strategy becomes visible. The builders of these monuments exhibited a profound understanding of their environment. The clusters are strategically positioned to overlook vital transit corridors (natural mountain passes) and are consistently located in proximity to perennial water sources (wadi beds and springs).

This specific positioning serves a dual function:

Pragmatic:

Proximity to water and pasture resources essential for agro-pastoral communities.

3.2 Symbolic/Defensive:

The occupation of high-visibility slopes implies a desire for "visual dominance" over the territory. In this context, the Bazinas and Dolmens acted as "territorial markers,"

physically asserting the tribe's claim to the land and its resources (Camps, 1961).

3.3 Socio-Spatial Organization:

The density of these sites in the rugged terrain of Zaccar and Djelfa, as opposed to the open plains, reflects a social organization deeply rooted in the highlands. The GIS analysis confirms that the "geography of death" (where people were buried) mirrors the "geography of life" (where they lived and moved). The alignment of these monuments suggests that Proto-historic tribes in the region were not aimless nomads, but possessed a structured sense of territoriality, using their ancestors' tombs to define the boundaries of their domain (Hachid, 2000).

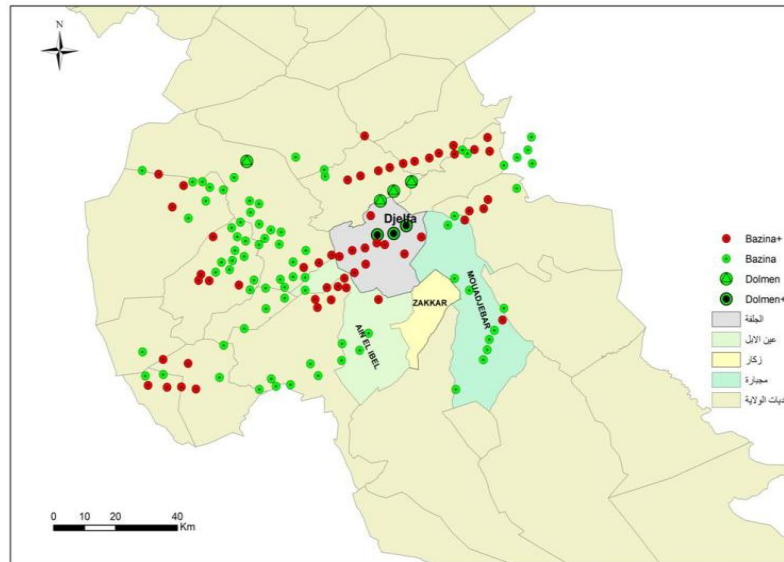


Figure 11: Spatial distribution of Bazinas and Dolmens in the Djelfa region. The map highlights the clustering of sites along the mountain slopes of Zaccar and Djelfa, illustrating the correlation between topography and funerary density.

4 Topographic Determinism and Settlement Ecology: A DEM-Based Analysis:

The application of Digital Elevation Models (DEM), as visualized in the appended Maps, allows us to move beyond simple 2D mapping to reconstruct the

three-dimensional reality of the Proto-historic landscape. The data reveals a profound correlation between site location and altitudinal zones, suggesting that the settlement patterns in the Djelfa region were governed by a strict form of "Topographic Determinism."

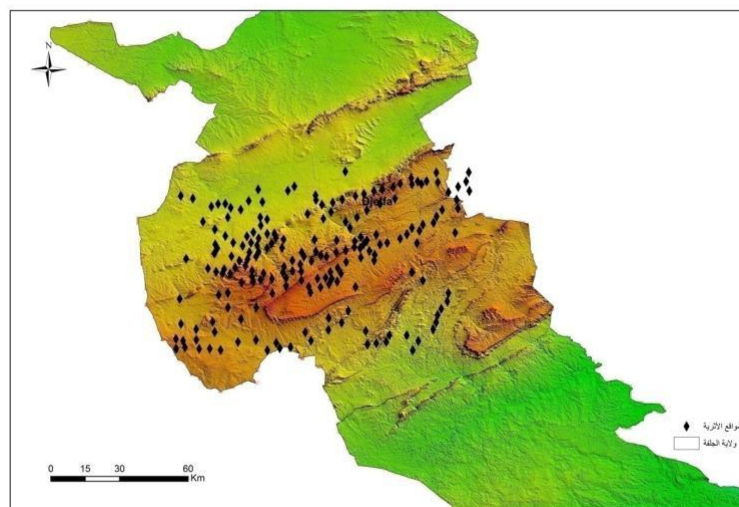


Figure 12

4.1 The Preferred Altitudinal Belt (1100m – 1400m):

Statistical analysis of the geo-referenced inventory indicates a deliberate selection of specific vertical niches. The vast majority of the archaeological sites recorded by Gsell are concentrated within a clearly defined altitudinal belt ranging between 1100 and 1400 meters above sea level. This specific range is not arbitrary; it represents an optimal habitation zone within the Central Saharan Atlas, balancing climatic comfort with resource accessibility

4.2 The Strategy of Avoidance and Adaptation:

By superimposing contour lines over the site distribution (Figure 13), we can reconstruct the environmental logic of these

ancient communities. The data suggests a sophisticated decision-making process characterized by a "strategy of avoidance":

Avoidance of Lowlands (< 1000m):

The scarcity of sites in the deep valley floors and alluvial plains indicates a conscious effort to mitigate hydrological risks, specifically the devastating flash floods (crues brutales) characteristic of the semi-arid wadi systems, as well as to avoid unhealthy, stagnant humidity.

Avoidance of High Peaks (> 1400m):

Conversely, the extreme summits were largely eschewed for permanent settlement, likely due to severe climatic exposure (high winds, freezing winter temperatures) and topographic inaccessibility.

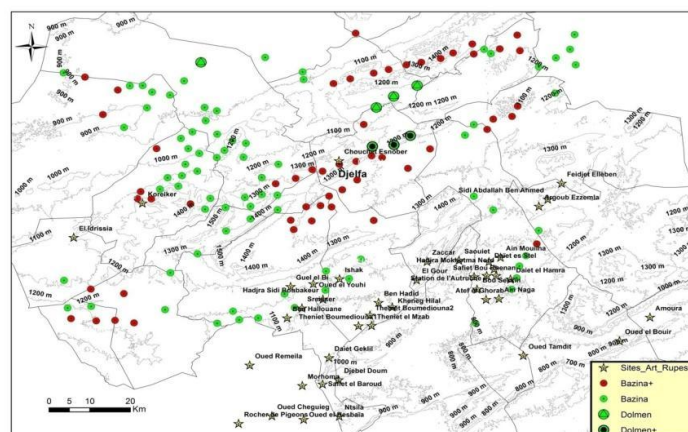


Figure 13: Rabhi Merouane's topographic Analysis Concept. This diagram illustrates how GIS visualizes terrain constraints. Similarly, this Map demonstrates how ancient populations selected the "middle ground"—avoiding both the flood-prone valleys (blue) and the harsh peaks (white/brown)—favoring the stable piedmont zones

4.3 The Piedmont Advantage: Defense and Subsistence:

Instead, the population concentrated in the "Piedmont Zones" (foothills) and mid-altitude terraces. This specific geomorphological placement offered a dual strategic advantage essential for survival:

Defensive Panopticism:

These elevated positions provided a superior "Viewshed" (field of view), allowing communities to monitor the surrounding plains and transit corridors effectively. In an era likely marked by inter-tribal competition, visual control over the territory was synonymous with security.

Ecotonal Resource Access:

Locating settlements in this transition zone placed these communities at the interface of two ecological niches: the fertile steppe suitable for grazing below, and the wooded uplands providing fuel and game above. This proximity allowed for a mixed agro-pastoral economy without sacrificing the defensibility of the site (Wheatley and Gillings, 2002).

5 Comparative Spatial Analysis: The Archaeological Atlas vs The Rock Art Archive:

Figure 14 stands as the synthesis of this entire study, offering a critical visual

juxtaposition between two distinct eras of archaeological inquiry. By overlaying the digitized inventory of Stéphane Gsell (represented in Red) with the modern inventory of rock art stations (represented in Green), the map functions as a palimpsest, revealing the layers of human history inscribed upon the Djelfa landscape.

5.1 Deconstructing the "Colonial Gaze" (The Red Points):

The spatial analysis reveals a clear methodological bias in Gsell's original work. The "Red" distribution is heavily skewed towards "Macroscopic Visibility". Gsell, and the French military correspondents of the early 20th century, prioritized structural remains—specifically Roman ruins, fortresses, and monumental Proto-historic tombs (Bazinas/Dolmens). Their survey was driven by what could be easily seen on the surface, often neglecting the more subtle, symbolic landscape of the deeper past.

5.2 The Prehistoric Substrate (The Green Points):

In contrast, the "Green" clusters representing rock art stations (engravings and paint-ings) occupy distinct geospatial niches. While there is some spatial overlap—indicating that certain locations held spiritual or strategic importance across millennia—the rock art extends into rugged sandstone ridges and shelters that Gsell's

atlas largely overlooked. These sites represent the Neolithic and Epipaleolithic substrate of the region, documenting a time

when the climate and fauna were radically different

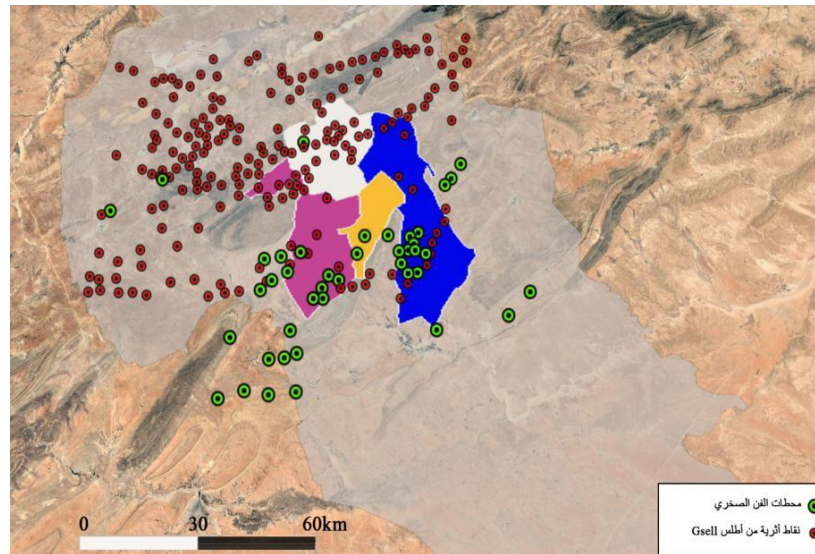


Figure 14: Rabhi Merouane

The Archaeological Reality. While Gsell documented the "stone structures" (left), modern research focuses on the "symbolic landscape" of rock art (Huard and Allard) (right). Figure 14 integrates these two worlds.

5.3 Diachronic Continuity:

The integration of these classical and modern datasets provides a holistic, diachronic view of the region's occupation. We can now trace the chronological trajectory of the Djelfa region:

Phase I (The Green Layer):

A landscape of hunter-gatherers and early pastoralists communicating through rock art (Prehistory).

Phase II (The Red Layer):

The shift towards monumentalism, territorial marking with stone tumuli, and eventually Roman fortification (Protohistory and Antiquity).

6 Conclusion and Future Perspectives

This research has demonstrated that the intersection of archival cartography and modern Geoinformatics constitutes a pivotal methodological advancement for North African archaeology. By subjecting Sheet No. 46 (Djelfa) of Stéphane Gsell's Atlas archéologique to a rigorous protocol of geometric rectification and spatial digitization, we have successfully bridged the "epistemological gap" between the static inventory of the early 20th century and the dynamic analytical requirements of the 21st century.

6.1 Methodological Validation:

Technically, the study proves that the obsolescence of colonial geodetic systems (Paris Meridian, Bonne Projection) is not an insurmountable barrier. Through the application of 2nd Order Polynomial Transformations and the use of high-precision control points (GCPs) derived from modern INC maps, historical errors were mitigated to statistically negligible levels. The resulting Spatial Database validates the accuracy of Gsell's original survey while enhancing it with the semantic richness and query capabilities of a modern GIS environment.

6.2 Reconstructing the Proto-historic Landscape:

Analytically, the transition from 2D maps to 3D Terrain Modeling (DEM) has fundamentally altered our understanding of the region's settlement dynamics. The identification of a distinct "Habitation Belt" between 1100m and 1400m confirms that the Protohistoric populations of Djelfa operated under a strict logic of Topographic Determinism. This settlement strategy was a calculated adaptation to the semi-arid environment, prioritizing the "Piedmont Zone" to balance the need for defensive surveillance (Viewshed) with access to hydraulic resources and steppe pastures.

6.3 The Diachronic Synthesis:

Furthermore, the comparative overlay of Gsell's "Red Points" (monumental architecture) and the modern "Green Points" (rock art) has generated a comprehensive, multi-layered map of the region. This synthesis corrects the "monumental bias" of early colonial archaeology, offering a more holistic view that encompasses both the symbolic landscapes of the Neolithic hunter-gatherers and the territorialized necropolises of the Protohistoric tribes.

6.4 Implications for Heritage Management (Preventive Archaeology):

Beyond its academic value, this digitized inventory serves as a crucial Spatial Decision Support System (SDSS) for heritage protection. In an era of rapid urbanization and infrastructural expansion in the Djelfa region, possessing a georeferenced map allows local authorities to integrate archaeological sensitivity zones into land-use planning (plans d'occupation des sols), moving from reactive salvage archaeology to proactive Preventive Archaeology.

7 Recommendations:

Consequently, this study advocates for the systematic generalization of this GIS-based protocol to the remaining 50 sheets of the Atlas archéologique de l'Algérie. The creation of a unified, national-scale Archaeological Geodatabase is no longer a luxury but an urgent necessity to preserve

the collective memory of the nation against the erosive forces of time and development.

References

[1] Camps, G. (1961). Monuments et rites funéraires protohistoriques: Aux origines

de la Berbérie. Paris: Arts et Métiers Graphiques.

Context in your article: Used to support the interpretation of the Bazinas and Dolmens, their typology, and their function as territorial markers during Protohistory.

[2] Conolly, J., and Lake, M. (2006). Geographical Information Systems in Archaeology. Cambridge: Cambridge University Press.

Context in your article: Referenced in the Methodology section to justify the treatment of physical map deformations (hygroscopic distortion) and the necessity of geometric rectification.

[3] Djellal, Y., and Belatra, R. (2019). "L'apport des SIG à la carte archéologique: Cas de l'Atlas Saharien". *Revue d'Archéologie Algérienne*.

Context in your article: A local, relevant study used to contextualize the application of GIS specifically within the Algerian Saharan Atlas environment.

[4] Gsell, S. (1911). Atlas archéologique de l'Algérie. Feuille N° 46 (Djelfa). Alger: Adolphe Jourdan.

Context in your article: The Primary Source. This is the document being updated, georeferenced, and analyzed throughout the study.

[5] Hachid, M. (2000). Les premiers Berbères: entre Méditerranée, Tassili et Nil.

Alger: Ina-Yas.

Context in your article: Used in the Discussion to link the spatial distribution of sites to the social organization of the first Berbers and to contrast the "stone monuments" with the "rock art" layer.

[6] Huard and Allard, Les figurations rupestres de la région de Djelfa

[7] Wheatley, D., and Gillings, M. (2002). Spatial Technology and Archaeology: The

Archaeological Applications of GIS. London: Taylor and Francis.

Context in your article: The theoretical backbone for the Spatial Analysis section. Used to explain concepts like "Viewsheds," "Cost Surface Analysis," and the transition from static maps to dynamic databases.